

Q1. Smartphone Performance Optimization

Problem

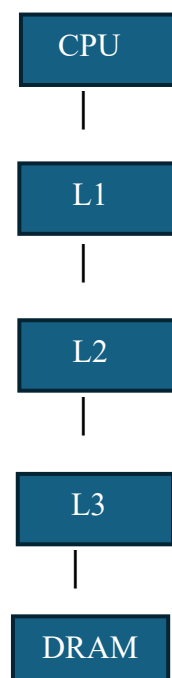
A smartphone manufacturer observes lag while switching between applications. The main reason can be inefficient memory access and insufficient cache utilization. The system uses **L1, L2, L3 cache and DRAM**, each having different latency and capacity characteristics.

Analysis

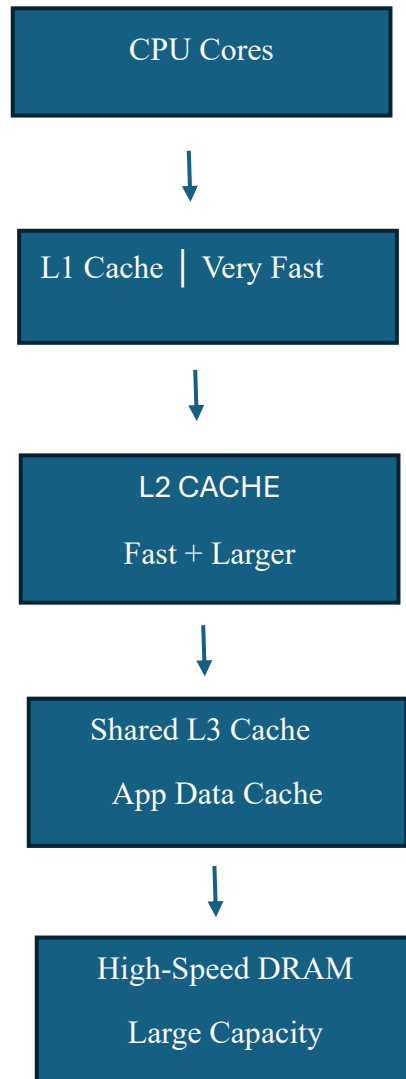
- **L1 Cache:** Smallest and fastest cache. It stores frequently used instructions and data.
- **L2 Cache:** Larger than L1 but slightly slower. It acts as an intermediate cache.
- **L3 Cache:** Larger and slower than L1 and L2. It can be shared among processor cores.
- **DRAM:** Provides large-capacity main memory but has much higher access latency than cache.

When switching between applications, frequently accessed data should remain in cache. If the required data is not available in the cache, the processor must access DRAM, increasing latency.

Proposed Memory Hierarchy



Proposed Improvements



Expected Result

The redesigned hierarchy reduces average memory access latency and improves application switching performance. Frequently used data is served from faster cache levels, while DRAM is used mainly for larger data requirements.

Conclusion

An efficient L1-L2-L3-DRAM hierarchy can reduce smartphone application-switching lag. Increasing cache effectiveness and reducing unnecessary DRAM accesses improves both latency and throughput.

Q2. Cloud Server Memory Bottleneck

Problem

A cloud service provider experiences high latency during database queries. The solution requires comparing **cache hierarchy and virtual memory paging** and selecting appropriate memory technologies.

Analysis

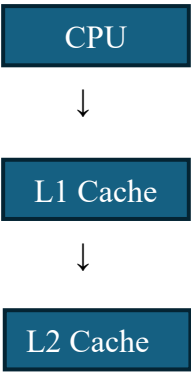
Database queries frequently access the same instructions and data. Cache memory can store frequently accessed database information close to the processor.

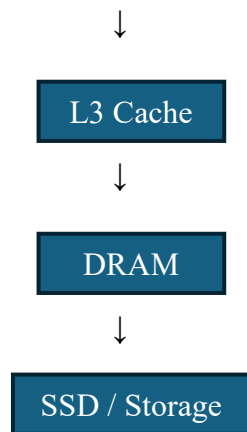
Virtual memory uses paging to provide a larger address space. However, when required data is not present in physical memory, a page fault occurs and data must be retrieved from secondary storage, causing high latency.

Cache vs Virtual Memory

Feature	Cache Hierarchy	Virtual Memory
Main purpose	Reduce memory access time	Extend available memory
Speed	Very high	Lower
Location	Close to CPU	Uses RAM and secondary storage
Latency	Very low on cache hit	High during page fault
Management	Hardware	Operating system
Effect on database	Improves query response	Provides larger memory space

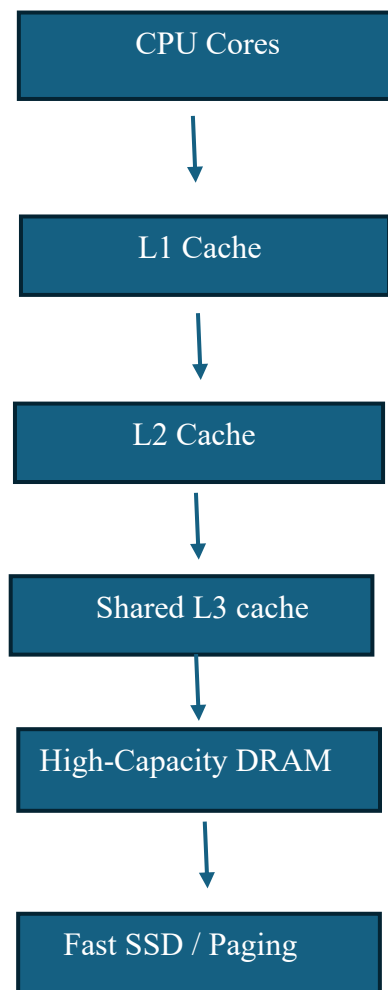
Recommended Solution





The server should use an effective cache hierarchy and sufficient DRAM to minimize page faults.

Improvements



Conclusion

Cache hierarchy should be prioritized for reducing CPU access latency, while sufficient DRAM should be provided to minimize virtual-memory page faults. This combination can reduce database-query latency and improve cloud-server throughput.

Q3. Autonomous Vehicle Data Processing

Problem

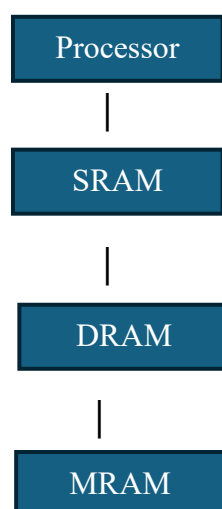
An autonomous vehicle must process sensor data in real time. The memory system must provide **high bandwidth and low latency**. The required technologies are SRAM, DRAM and MRAM.

Analysis

Autonomous vehicles continuously process data from cameras, radar, LiDAR and other sensors. Delays in processing can affect real-time decision making.

- **SRAM:** Very fast and suitable for frequently accessed sensor-processing data.
- **DRAM:** Provides larger capacity and is suitable for storing larger working datasets.
- **MRAM:** Non-volatile and provides fast access with good energy efficiency.

Proposed Hierarchy



Memory Allocation

Memory	Suggested Use
SRAM	Critical real-time data and frequently accessed instructions
DRAM	Large sensor datasets and active applications
MRAM	Persistent/critical data requiring fast access

Improvements

Sensor Processing

Camera / Radar Data

SRAM Cache | Ultra-Low | Latency

High-Speed | MRAM | Fast Access

DRAM | Large Working | Memory

Expected Result

Frequently accessed and time-critical information can be accessed from SRAM with minimum latency, while DRAM provides the required capacity and bandwidth. MRAM provides persistent storage with fast access.

Conclusion

A combined SRAM-DRAM-MRAM hierarchy provides a balance between **latency, bandwidth, capacity and energy efficiency**, making it suitable for real-time autonomous vehicle data processing.

Q4. AI Inference Edge Device

Problem

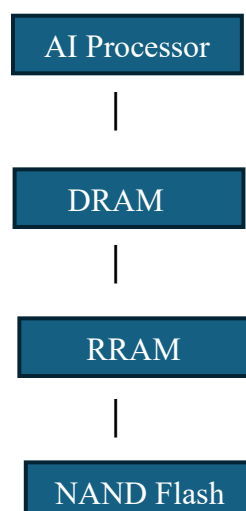
An edge AI device requires fast AI-model loading while maintaining low power consumption. The required technologies are NAND Flash, DRAM and RRAM.

Analysis

AI models can be large and require fast access during inference.

- **NAND Flash:** Non-volatile and suitable for permanent storage of AI models.
- **DRAM:** Fast working memory for actively running AI models.
- **RRAM:** Non-volatile technology with potential for fast and energy-efficient memory operations.

Proposed Hierarchy



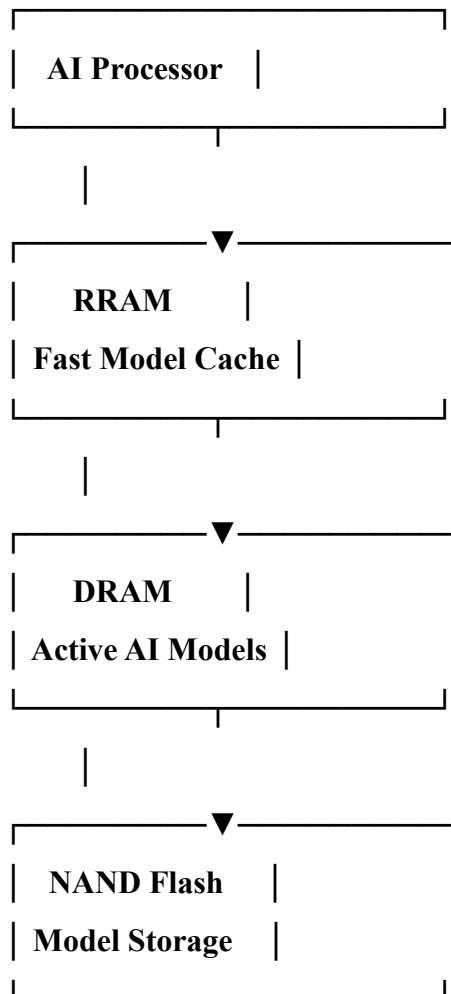
Memory Usage

NAND Flash: Stores the complete AI model permanently.

RRAM: Can store frequently used model parameters or selected data with low-power operation.

DRAM: Loads the active model and intermediate data during inference.

Improvements



Expected Result

The system reduces model-loading time by keeping frequently required information in faster memory and reduces energy consumption by minimizing unnecessary data transfers.

Conclusion

A combination of NAND Flash, RRAM and DRAM provides a suitable hierarchy for edge AI devices by balancing **storage capacity, access speed and power efficiency**.

Q5. High-Performance Gaming Console

Problem

A gaming console experiences frame drops while loading large scenes. The problem requires analysis of **L3 cache and DRAM throughput** and improvements to the memory hierarchy.

Analysis

Large game scenes contain textures, models, animations and other data. During scene loading, large amounts of data must be transferred between storage, DRAM, cache and the processor.

L3 Cache

L3 cache is faster than DRAM and can provide frequently accessed data to processor cores with lower latency. However, its capacity is limited compared with DRAM.

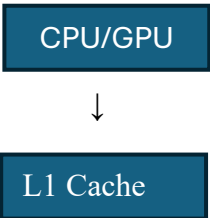
DRAM

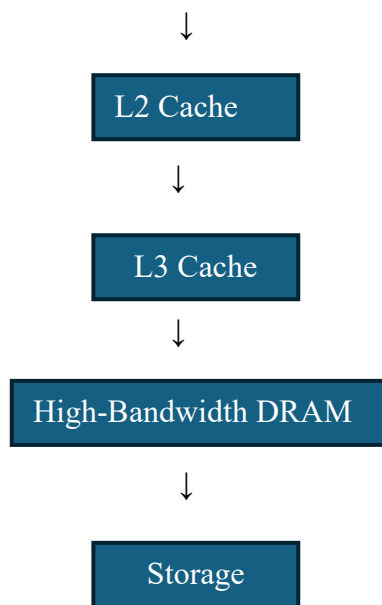
DRAM provides much larger capacity and is suitable for storing active game assets. However, it has higher latency than cache.

Comparison

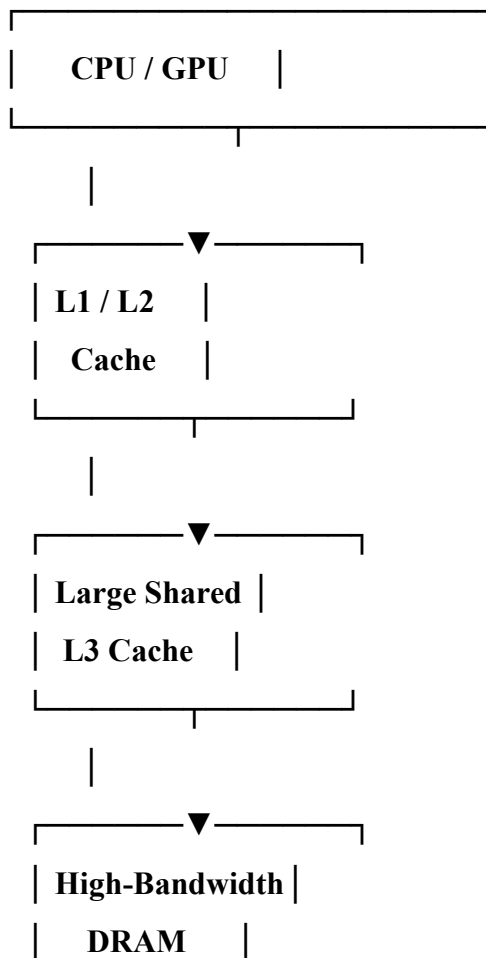
Feature	L3 Cache	DRAM
Latency	Lower	Higher
Capacity	Smaller	Larger
Speed	Faster	Slower
Purpose	Frequently accessed data	Large working datasets
Processor proximity	Very close	Further from CPU

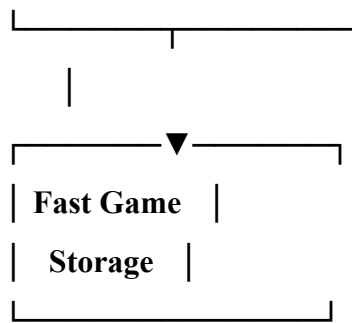
Proposed Hierarchy





Improvements





Expected Result

Frequently accessed game data can be supplied quickly from cache, while high-bandwidth DRAM handles large scene data. This reduces memory bottlenecks and helps maintain stable frame rates.

Conclusion

Combining an efficient L3 cache with high-bandwidth DRAM can reduce scene-loading delays and minimize frame drops. A well-designed memory hierarchy improves both latency and throughput for gaming workloads.

These solutions directly address the CO4 requirement of designing hierarchical memory systems using **L1, L2, L3, DRAM, SRAM, NAND Flash, MRAM and RRAM** based on latency, bandwidth and throughput.